Prasenjeet Tambe

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Education

University of Southern California (USC), Viterbi School of Engineering — **Los Angeles, CA**
Master of Science in Computer Science — *Aug 2026 – Dec 2028*

Pune Institute of Computer Technology, Savitribai Phule Pune University — **Pune, India**
B.E. in Electronics and Telecommunication Engineering; CGPA: 8.76/10 — *Aug 2019 – Jun 2023*

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, C++, SQL, HTML/CSS
Frameworks & Libraries: ReactJS, React Flow, Node.js, NestJS, Flask, LangChain, GraphQL, Prisma ORM, SQLAlchemy, TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, Tesseract, EasyOCR, Granite Docling
Databases: PostgreSQL, MySQL, MongoDB, Firestore, Redis
Cloud & DevOps: AWS (ECS, EC2, RDS, Lambda, S3, SQS, Step Functions, EventBridge, API Gateway, CI/CD), GCP (App Engine, Pub/Sub, GCS, Firestore, Cloud Run), Nebius (H200), Git, GitHub, GitLab, Docker, GitHub Actions

Experience

Insaito — **Remote**
Senior Software Engineer — *Sep 2025 – Aug 2026*

- **Founding Engineer** who architected and built **Automatan.ai**, an end-to-end **AI Agent platform**, owning frontend, backend, AI workflows, and cloud infrastructure using **React, Python, Flask, and GCP**.
- Implemented **multi-agent orchestration** to coordinate agents and execution flow within the platform's AI workflows.
- Built **retrieval-augmented generation (RAG) pipelines** integrating document processing and **vector databases** to index and retrieve unstructured content, enabling semantic search and grounding **LLM responses** in source documents.
- Developed a **document intelligence pipeline** using **Docling, OCR, and VLMs**, processing **1,000+ documents daily** to extract **user-selected insights** as structured **JSON**; evaluated **LLM extraction** against verified ground-truth examples using task-specific metrics, achieving **95%+ extraction accuracy**.
- Engineered an event-driven **Google Sheets automation engine** processing **10,000+ sheet updates**, triggering AI transformations via **Pub/Sub**, persisting results in **Firestore and GCS**, and displaying real-time processing status.

Fold Health Inc. — **Pune, India**
Software Engineer — *Jul 2023 – Jun 2025*

- Designed and maintained a **Workflow Automation System** used by **200+ healthcare providers**, supporting **1.7M+ active lives across 10+ B2B healthcare organizations in the US**, saving 75% time with 40+ triggers and 100+ filters using React Flow, ReactJS, AWS Lambda, AWS Serverless Application Model, Node.js, Python, and PostgreSQL.
- Reduced **API calls by 3x** by optimizing **ReactJS** data fetching with **useEffect** dependency management, memoizing components and callbacks with **React.memo**, **useMemo**, and **useCallback**, and restructuring **Context API** state to minimize re-renders; implemented **Redis caching** and query optimizations to reduce backend load and latency.
- Led the end-to-end migration of existing workflows to a **scalable architecture**, increasing system throughput from **20 to 200 concurrent workflows (10x)** while maintaining compatibility with existing workflows.
- Built a serverless PDF pipeline on **AWS Lambda**, supporting **1,000+ concurrent requests**; implemented database backed caching to reuse documents, eliminate redundant invocations, and reduce compute overhead.
- Architected and delivered an **LLM-powered EHR synopsis** (Flask, LangChain, PostgreSQL), owning backend, database design, template UI, and patient-profile integration with **HIPAA compliance** and **role-based access controls**.
- Reduced **LLM inference costs by up to 80%** through **prompt optimization** and **response caching** to eliminate redundant invocations; implemented **output normalization** to standardize structured responses for frontend consumption.
- Built and shipped **50+ production features** across healthcare products, collaborating with senior architects, designers, and the product team; contributed to **code reviews**, end-to-end testing, releases, and production issue resolution.

Projects

Pulse Rate Monitoring | *OpenCV, Signal Processing* — **B.E. Final-Year Project**

- Developed a contactless pulse monitoring system using **OpenCV**, processing webcam and external camera recordings to track pixel intensity variations within a **region of interest (ROI)** and construct a time-series signal for pulse estimation.
- Applied **bandpass filtering** to suppress noise outside the pulse-related frequency range and **Fourier Transform** analysis to identify the dominant frequency and calculate BPM, addressing lighting variations and movement during signal extraction.

Achievements & Certifications

- **National Rank 6** (2022) and **National Rank 13** (2020), ABU Robocon Asia-Pacific (India); Core Technical Member, Pune Institute of Computer Technology Robocon Team (2019-2023)
- ML for All – University of London; Heterogeneous Parallel Programming – AstroMediComp.org
- **Coursework:** Database Systems (CSCI 585), Introduction to Programming Systems Design (CSCI 455x), Data Structures, Object-Oriented Programming, Computer Networks, Cloud Computing, Deep Learning